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CS 341 Software Engineering
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Sprint 3

This is a team assignment, your group can (and should) divide up the work. Please read this whole document carefully before getting to work.

As usual, the sprint lead for Sprint 2 needs to send an assessment email to each member of their team awarding the 0-N tokens that are proportional to their performance during Sprint 2. The email has to contain feedback on their performance (what they did well, what they need to improvement and any other feedback you see fit). Don't hesitate to include your instructor on the email.

YOU ARE ALLOWED TO DIVERGE FROM YOUR VISION OR SRS!

In Trello: Sprint 3 Board

1. Sprint planning
2. Tracking progress

MVP!

You need the basic functionality for this sprint:

- Load & display real data from at least one database
- Migrate your data to database and include all necessary CRUD forms? **CREATE**, **READ**, **UPDATE** and **DELETE**
- Data visualization or analysis
- UI is completely prototyped (all pages navigable, menus, button, images all laid out)
- Test functions for each feature.
- Graceful error handling

Travis-CI and Codecov!

1. Add to your git ignore file for the following directory: `/coverage`
2. Go to `travis-ci.com` and log in with your github account. Be sure to use your free access from the Student Pack. (Warning: `travis-ci.org` will not work even though it looks the same.)
3. (Only if prompted, Click request access for the upcs organization and in the modal dialog click request. May be up to TWO requests).
4. (Click Authorize if you have to.)
5. On Travis CI: Enable your project repository listed under the upcs organization.
6. Get Travis CI to link to your tests and successfully complete a test build. (You will need to add a Travis CI file to your github, you will also need to read the how to documentation on Travis CI. **Documentation->Languages->JavaScript has what you need.** You should make sure that you are using node_js version 10 or newer!)
7. Code coverage. Add to your Travis CI file information for code coverage. Use this service: <https://codecov.io> Here is a handy link: <https://docs.codecov.io/docs/supported-languages>
 - Hints: You need to install this with npm and you need to add that install line to your .yml file. You also need to update your package.json with codecov and a token
 - You can see your output after npm test here `/coverage/lcov-report/index.html`
8. Finally add a code coverage badge to your README.md. Modify this to get it to show up.
[[Codecov Coverage](<https://img.shields.io/codecov/c/github/<GithubUsername><RepositoryName>/<BranchName>.svg?style=flat-square>)]
(<https://codecov.io/gh/<Github Username>/<Repository Name>>/)

9. **Submit** a screenshot pdf of your code coverage from any point during this sprint on moodle. And a screenshot of a successful build on Travis-CI (show passing tests). Only one team member should submit.
10. Make sure jest or other unit testing framework is setup and working.
11. Exception: if your project uses different code coverage framework, feel free to use it.

Develop! Beta Version

Functional completeness is expected for a Beta Version. Your primary features should be finished. You need the basic functionality for this sprint:

- A project VM server setup to run your node server. For now you can manually copy code from github using the clone command to get your latest code onto this server.
- Ideally this should be as simple as logging into your server and, git cloning your github project and running `npm start`. *However, the bonus here is to create a startup script so that your instance automatically restarts the server on a reboot of any kind. Or if your application crashes (make sure a crash report is logged and can be used to troubleshoot your app).*
- Live data must be able to be displayed in at least two different ways (filter the data)
- Additional major functionality based on team goals
- Test functions for each feature.
- Graceful error handling (i.e. mismatched tags shouldn't break your beautiful display)
- Continued focus on your team's project goals
- **This should be a nearly complete project at the end of this sprint in terms of your Minimum Viable Product (MVP)**

Ideas for next steps (don't forget to look at previous sprint suggestions)

- Improved security (such as SSL)
- Protect router endpoints against exposed API
- Excellent visuals
- Ability to change the website theme or data colors/shapes
- User login
- Data collection forms (i.e. sign up for an account)
- Find & Import new data into a database
- Other interesting or valuable features?

Reflect (Review & Retrospective)

Before the last day of the sprint you should reflect:

- What went well during the sprint cycle?
- What went wrong during the sprint cycle?
- What could we do differently to improve?

In class we will review, which may include code review, demos, and comments on the reflections.

As part of your reflection, you may want to take a moment to email the instructor with any comments you have so far regarding this sprint or the class as a whole.

Submission

As the halfway point in this project, you should fill out a peer evaluation before you leave for spring break or right after you come back. You will get participation points for completion.

The "submission" of your code will be the most recent version of your software (on Github) prior the start of class on the last day of the sprint. Your instructor will check to see that Travis-CI is in a

successful build state, your code may be cloned by your instructor and tested for the functionality requested. It is highly recommended that you include a summary of completed features (and if necessary how to use them), in your top level README file

Grading

Multiple git pushes by all team members.

Minimum requested features implemented including test functions

Additional features implemented as time permitted including test functions

Correct and complete use of Trello

Travis-CI & code coverage integration is active

Rubric

Features (& tools setup) as specified in sprint 35%

Additional features as defined by team. 20%

Git commits, Trello use 25%

Complete Testing and Coverage 15%

Teamwork 5%

The entire grade for all team assignments may be negatively modified for individuals who display unsatisfactory participation, attendance, or teamwork.